

- 17** High concentrations of ethanol can disrupt cell membrane structure and make the cell surface membrane leaky.

Yeast cells release ethanol as a waste product of metabolism.

Yeast cells can alter the composition of the cell surface membrane to reduce the effect of ethanol.

- A saturated fatty acid, X, is replaced by an unsaturated fatty acid, Y.
- Y has a shorter hydrocarbon chain than X.
- There is an increase in the phospholipid to protein ratio.

Which statement would explain why the effect of ethanol is reduced?

- A** A greater proportion of phospholipid may increase the entry of ethanol via channel proteins.
- B** An increase in hydrophobic interaction by phospholipid tails improves membrane stability.
- C** An increase in the proportion of fatty acids with double bonds decreases membrane fluidity.
- D** The shorter hydrocarbon chain of Y decreases membrane fluidity.